

CSMFAB000000000113-Vol-22

Launch Platform: Vessel Design and Multi-Boat Coordination System

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1. Vessel Specifications for AMSR-UBSCLS Operations

1.1 Master Vessel (Command Boat)

The master vessel performs: - Launch sequencing and LoRa command broadcast - Spear deployment and underwater positioning - Recovery coordination post-launch

Requirements: - Length: 10-15 m (30-50 ft) — sufficient for rope handling + dive team - Hull: BFRP/Elum® (GIC-immune, CSMFAB00000000011-compatible) - Propulsion: diesel-electric hybrid — main engine stopped during launch to reduce vibration - Crew: 4 minimum (pilot, dive coordinator, electronics operator, safety officer) - Equipment: BFRP davit crane for spear deployment, rope spool, LoRa master station

Standard research vessel options (charter or purchase): - Rigid Inflatable Boat (RIB) 12m with BFRP hull modification: ~\$85,000 - Purpose-built BFRP research vessel 12m: ~\$180,000

1.2 Slave Vessels (n-1 Support Boats)

For n=6 configuration: 5 slave vessels positioned at hexagonal ring.

Minimum slave vessel requirements: - Length: 5-8m (sufficient to carry 50m rope spool + actuator) - Hull: fiberglass acceptable (GIC-immune alternative: BFRP) - Crew: 2 per vessel minimum - Equipment: rope spool BFRP frame, LoRa slave radio, ZrO2 ceramic cleat for rope attachment

Budget options for hobbyist scale (n=4, 10m ropes): - Inflatable kayaks with BFRP frame insert: ~\$800 each - Small aluminum motorboats: acceptable (far from spear center — minimal GIC risk at 6m radius) - Total slave vessel budget for 3 support boats: \$2,400 minimal

1.3 Vessel Positioning Protocol (GPS-Free)

LoRa ranging system: 1. Master vessel: LoRa broadcast position reference (ranging beacon mode) 2. Each slave vessel: measures time-of-flight to master → computes distance 3. Slave-to-slave ranging confirms hexagonal geometry within $\pm 0.5\text{m}$ 4. All vessels motor to calculated position

Positioning accuracy required: $\pm 2\text{ m}$ (from Vol 05 analysis) LoRa ranging accuracy: $\pm 0.5\text{ m}$ — **meets requirement.**

No GPS required. Entire system operates in GPS-denied Carrington-event scenario.

2. Rope Deployment Sequence

Step 1: Anchor Spear Beneath Vessels

- Pneumatic guide weight (BFRP housing, N2-filled) attaches to spear base
- Guide weight drops spear to target depth (30m for Scale 3)
- Time: 60 seconds descent

Step 2: Vessel Circle Formation

- Master vessel marks center; all boats motor to hexagonal positions
- LoRa ranging confirms geometry within $\pm 1\text{m}$
- Time: 15-20 minutes

Step 3: Rope Connection

- Diver from each vessel descends to 2m depth
- Clip Smart Rope terminal connector (ZrO2 bayonet) to spear base lug
- Surface; rope naturally hangs upward (positively buoyant)
- Time: 5 minutes per rope; all 6 in parallel = 5 minutes total

Step 4: Pre-Tension

- Each vessel's winch takes up rope slack
- Tension load cell (FBG optical) confirms 5000 N pre-tension per rope
- Time: 2 minutes

Step 5: Final Checks

- Electronics team: confirms all LoRa nodes active
- Spear spin-up: verified via angular rate sensor reading (FBG optical)
- Sea state: confirmed Beaufort 3
- Safety zone: 500m radius confirmed clear by radar sweep

Step 6: Launch Sequence

- Master vessel transmits ARMED command
- 30 second countdown broadcast via LoRa
- FIRE command: all 6 rope actuators trigger simultaneously
- Time: < 0.1 ms from command to all-rope actuation

Total preparation time (calm seas): **45-60 minutes**

3. Recovery Protocol

Post-launch, Smart Ropes float to surface (positive buoyancy). Recovery: 1. Each vessel retrieves its rope (3-5 minutes per rope) 2. Spear recovery: GPS beacon (or LoRa on spear) tracked as it descends after apogee 3. Spear lands ~50-250 km downrange at sea 4. Recovery vessel (fast RIB) dispatched to coordinate: ~1-3 hour recovery

For suborbital missions: Spear enters at high velocity — recovery via parachute deployed by KNbO₃-BaTiO₃ mortar at atmospheric entry, or intentional ocean impact (expendable missions).

Citations (Vol 22)

- CSMFAB000000000011 V2.0, BFRP vessel hull design, CSM 2026
- LoRa Alliance, LoRaWAN Ranging Protocol, 2024
- Bowditch, N., American Practical Navigator, NGA 2019
- USCG, "Operating Requirements for Small Vessels in Offshore Waters," CFR Title 46, 2024

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